

Defining a microservice architecture

Marigold Hotels example

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Chapter 1. Introduction

In Chris Richardson's *Microservice Architecture definition workshop*, you will define a microservices-based architecture for a hotel management application.

The starting point is the application's requirements.

Chapter 2. Hotel management application requirements

Here the requirements for the hotel management application.

2.1. Vision

Build a hotel management application for Marigold Hotels, which is a hotel chain with numerous properties around the world.

2.2. System Context



Customer:

- makes reservations and stays at hotels

Hotel operator:

- Defines properties and pricing

Front desk staff:

- Checks in and out guests

2.3. Stories

2.3.1. Property management

There are several property management stories.

Story: Create Property

Feature: Create property

As a hotel operator
I should be able to create a property
So that I can start accepting guests

Scenario: Property created
Given Some valid room types
When I create a property with an address of (123 Some Stret, Anytown, USA) with three rooms (101 ...)
Then the property is created

Story: Create Room type

Feature: Create room type

As a hotel operator
I should be able to create a room type
So that I can start accepting guests

Scenario: room type created
Given there isn't an existing room type of XYZ
When I create a room type of XYZ
Then the room type is created

Story: Set prices

Feature: Set prices

As a hotel operator
I should be able set the price chart for a room type on a given day
So that I can start accepting reservations

Scenario: set prices
Given A valid property
Given a date in the future
Given a room type for that property
Given a list of valid (room rate, quantity) pairs
When I set prices
Then the price chart [(quantity, room rate), ...] for that (property, room type, date) is created

The big idea is that on a given day and room type, there are different price buckets. Each bucket contains some number of rooms at a particular room rate. A room type is offered at the lowest pricing bucket until that bucket sells out. The room is then offered at the next lowest pricing bucket.

Here is an example of pricing for rooms with a king bed on a given day:

Number of rooms	Rate
10	\$150
20	\$180
20	\$220
10	\$300

The first 10 rooms are sold at a very low rate. The next 20 at a slightly higher rate. And so on.

Note: in practice, a hotel will use a dynamic pricing that periodically calculates the room pricing. However, the mechanism that calculates prices is outside the scope of this exercise.

2.3.2. Reservations

There are several reservation management stories.

Story: Make reservation

Feature: Make reservation

As a customer

I should be able to make a reservation

So that I can stay at a hotel

Scenario: make a reservation

Given a valid credit card

Given a stay date range that is not in the past

Given a valid property

Given a room type for that property that is available for all of the stay dates

Given available daily room rates for that room type for each the days of the stay

When I make a reservation

Then a reservation is created for the stay date range and the displayed rates

Then the availability of that room type for the stay range is reduced by one

Note: the reservation process is as follows:

1. The customer enters an address and the stay date range.
2. The system displays properties with available rooms.
3. The customer selects a property
4. The system displays the available room types and rates.
5. The customer selects a room type/room rate and enters their details including the credit card
6. The system creates a reservation

Key ideas:

- A room type is available if for each day of the stay the number of rooms of that type > number of reservations on that day for that room type
- The available room rates for a given day are determined as follows:
 - Count the number of reservations for that day for that room type
 - Sort the property's DailyRoomRates by increasing price
 - Find the first DailyRoomRate that has not be 'used up'

2.3.3. Front desk

There are several front desk stories.

Story: Check in

Feature: Check in

As a front desk employee
I should be able to check in a guest
So that the guest can begin their stay

The flow is as follows:

1. Front desk agent searches for reservations
2. System displays reservation
3. Front desk agent enters credit card details
4. System displays available rooms
5. Front desk agent selects room
6. System checks in guest

Scenario: guest check in

Given A guest with a valid reservation
Given using a valid credit card
Given and a room of the reserved type is available
When I check in the guest
Then the room is assigned to the guest
And the state of the guest is checked in

Story: Check out

Feature: Check out

As a front desk employee
I should be able to check out a guest
So that the guest can leave the hotel

Scenario: guest check out

Given A guest that is checked in
Given using a valid credit card
When I check out the guest

Then the state of the room is available
And the state of the guest is checked out
And the guest's credit card is charged

2.4. Example user experience for making a reservation

From Hilton.com

2.4.1. Enter location and stay dates

The screenshot shows the Hilton.com reservation interface. At the top, the Hilton logo is on the left, and navigation links for Locations, Offers, Meetings & Events, and Credit Cards are in the center. On the right, there are links for Join and Sign In. Below the navigation bar, the 'Where to?' field contains 'London, UK'. To the right of this field, the selected dates are '14 SEP WED' and '21 SEP WED', with a '1 Room, 1 Guest' button and a 'Special Rates' button. A 'Find a Hotel' button is on the far right. A calendar modal is open, displaying August 2022 and September 2022. The calendar shows the dates 14 and 17 for August, and 14 and 21 for September. The date 21 is highlighted. At the bottom of the calendar, there is a checkbox for 'My Dates are Flexible' and a 'Close' button.

Hilton
FOR THE STAY

Locations Offers Meetings & Events Credit Cards

Join Sign In

Where to?
London, UK

14 SEP WED 21 SEP WED 1 Room, 1 Guest Special Rates Find a Hotel

Join Hilton Honors and get up to 10% off your stay

for Free

August 2022 September 2022

SUN MON TUE WED THU FRI SAT SUN MON TUE WED THU FRI SAT

1 2 3 4 5 6 1 2 3

7 8 9 10 11 12 13 4 5 6 7 8 9 10

14 15 16 17 18 19 20 11 12 13 14 15 16 17

21 22 23 24 25 26 27 18 19 20 21 22 23 24

28 29 30 31 25 26 27 28 29 30

☐ My Dates are Flexible Close

2.4.2. Pick available property



Edit Stay Details: London, UK, 14 Sep - 21 Sep

Sort by

Distance

Select Currency ^①

Hotel

☐ Use Points

Filters

Showing 1 - 20 of 115 hotels

*Price change during stay



The Trafalgar St. James
London, Curio Collection by Hilton

0.08 miles

[Hotel Details >](#)

Rating: 4.5 out of 5.0



Based on 1037 Reviews >

£462

HONORS STAY LONGER AND SAVE

[View Rates](#)



The Waldorf Hilton, London

0.50 miles

[Hotel Details >](#)

Rating: 4.0 out of 5.0

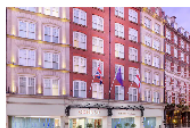


Based on 5899 Reviews >

£408*

HONORS STAY LONGER AND SAVE

[View Rates](#)



Conrad London St. James

0.61 miles

[Hotel Details >](#)

Rating: 4.5 out of 5.0

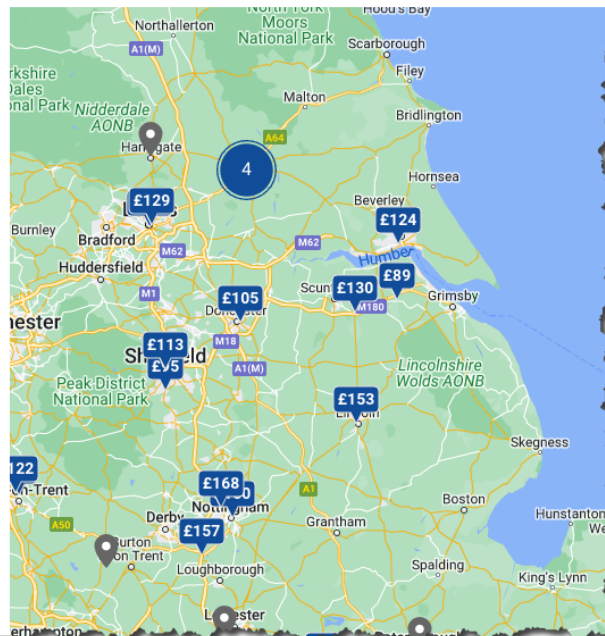


Based on 5331 Reviews >

£390

HONORS STAY LONGER AND SAVE

[View Rates](#)



2.4.3. Select room type



Join | Sign In

What To Expect During Your Stay [Learn More](#)

Your Stay Wed, Sep 14 – Wed, Sep 21, 2022 (7 nights) 1 room for 1 adult [Edit Stay](#)

Step 1 of 3

Select a Room

Your stay includes

✓ Non-smoking rooms ✓ On-site restaurant ✓ Indoor pool ✓ Fitness center ✓ Business center

Room Filters

Special Rates



Connecting Rooms



Use Points & Money



The Waldorf Hilton, London

Aldwych London WC2B 4DD United Kingdom

[Hotel details >](#)

On school or national holidays, this hotel offers a bespoke Family Experience for travellers below the age of 12. Children enjoy a Kids Welcome and a dedicated Kids Breakfast experience. Enjoy your family meals by selecting dishes from the Kids Menu. The hotel has kids amenities available on request

We found 9 rooms for you.

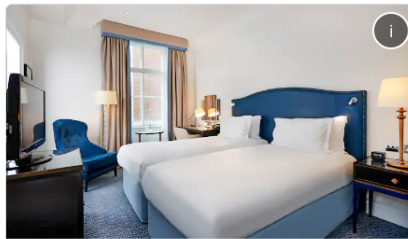
Avg/night including tax in £GBP

***Price change during stay**



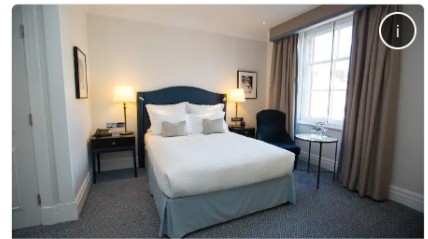
Queen Superior Room

Book From £408*



Twin Deluxe Room

Book From £416*



Queen Deluxe Room

Book From £416*

2.4.5. Enter guest and payment information



Your Stay  Wed, Sep 14 – Wed, Sep 21, 2022 (7 nights)

 1 room for 1 adult

[Edit Stay](#)

Step 3 of 3

Payment and Guest Details

Total for stay

£3,304.07

[Show price details](#)

Total room charge

£3,304.07

Total taxes

Included in price

 This is a members-only rate. Join Hilton Honors (for free!) on this page, [sign in](#), or [choose a different rate](#).

All fields are required unless marked optional.

Payment

Card number

First name on card

Last name on card

Month



Year



The Waldorf Hilton, London
Aldwych London WC2B 4DD UK
[Hotel details >](#)

 [Queen Superior Room](#)
Change Room

 [Honors Discount](#)
Change Rate

 [Payment and Guest Details](#)

[View optional services](#) ▼

Guarantee and Cancellation Policy

There is a credit card required for this reservation. Free cancellation before 11:59 PM local hotel time on 13 Sep 2022.

By clicking "Book Reservation," I agree to the [Rules and Restrictions](#), [Site Usage Agreement](#) , and agree that Hilton will collect, use, share and transfer my information as set out in [Hilton's Global Privacy Statement](#) .

California consumers, [click here](#)  to learn about Hilton's collection and use of your personal information.

I also agree to the [Hilton Honors Program Terms and Conditions](#) .

Book Reservation

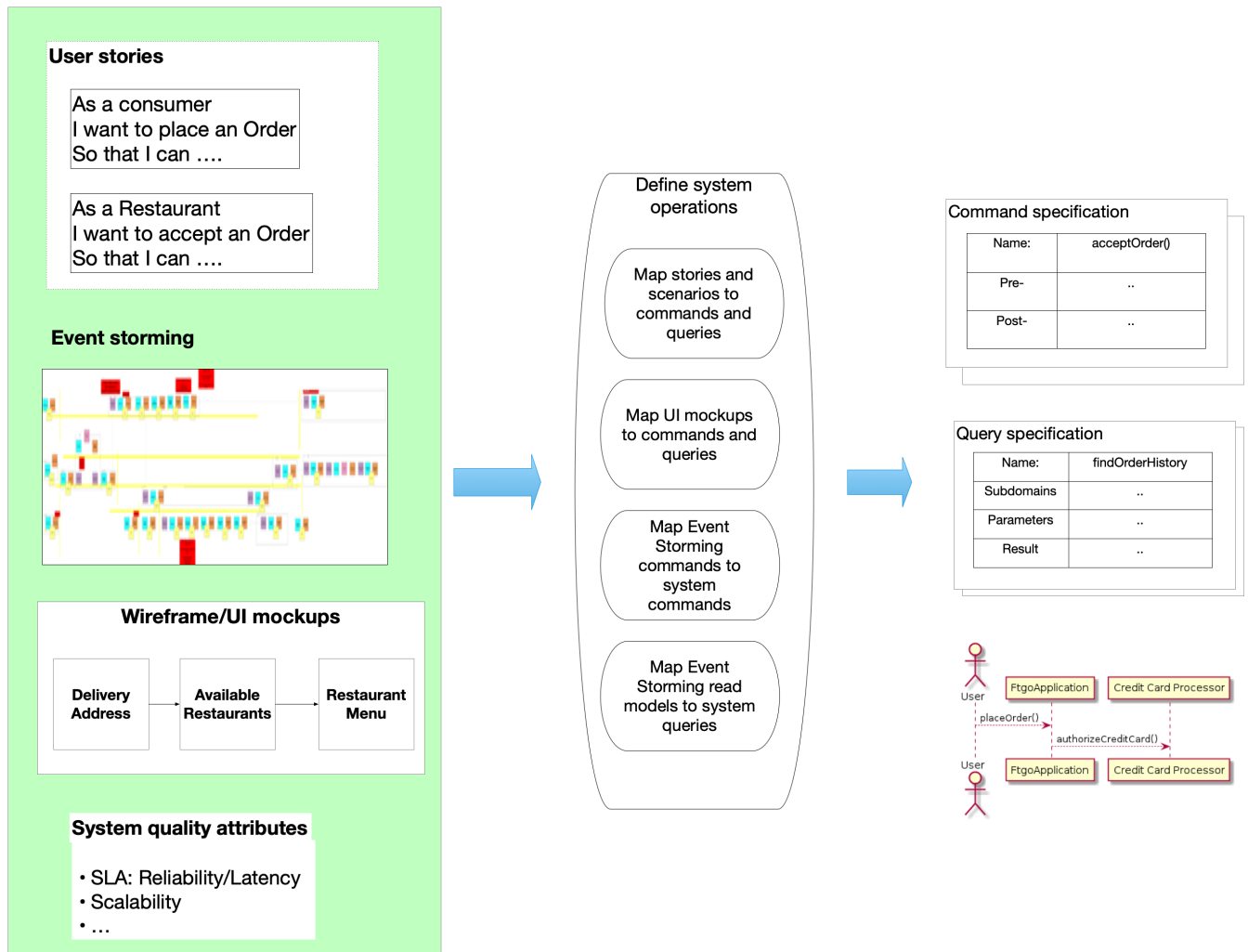
2.4.6. Display reservation conformation

NOT CAPTURED:

- **Reservation confirmed** page - shows the reservation number etc

Chapter 3. Kata #1: Discover system operations

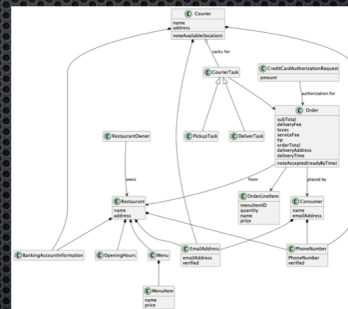
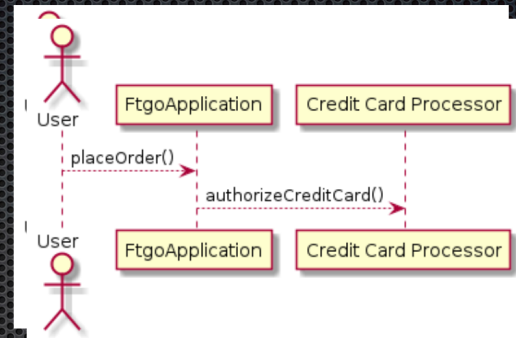
Define and document the system operations using the templates in [01-discovering-system-operations](#)



3.1. Deliverables

Summary: deliverables

System command canvas	
Name	<code>placeOrder(consumerID, deliveryTime, deliveryAddress, restaurantID, lineItems)</code>
Parameters	• <code>consumerID</code> - the ID of the consumer placing the order • <code>deliveryTime</code> - the desired delivery time • <code>deliveryAddress</code> - the delivery address • <code>restaurantID</code> - the restaurant • <code>lineItems</code> - collection of <code>OrderLineItem(menuItemIDs, quantity)</code>
Return value	• <code>orderID</code>
Trigger	User command
Service Objectives	Level • Business tier: Critical, Core, User, Internal • Throughput, e.g. requests/second • Availability - ... • Latency - ... • ...
Preconditions	• a Consumer with ID <code>consumerID</code> exists • the Consumer has a valid payment method • a Restaurant with ID <code>restaurantID</code> exists • the <code>deliveryAddress</code> and <code>deliveryTime</code> are served by the restaurant • <code>MenuItem</code>s identified by <code>menuItemID</code> exists
Postconditions	• an <code>Order</code> order was created in the <code>APPROVED</code> state • a <code>CreditCardAuthorization</code> for the <code>orderTotal</code> for the Consumer's credit card was created • <code>returnValue</code> became <code>orderID</code>



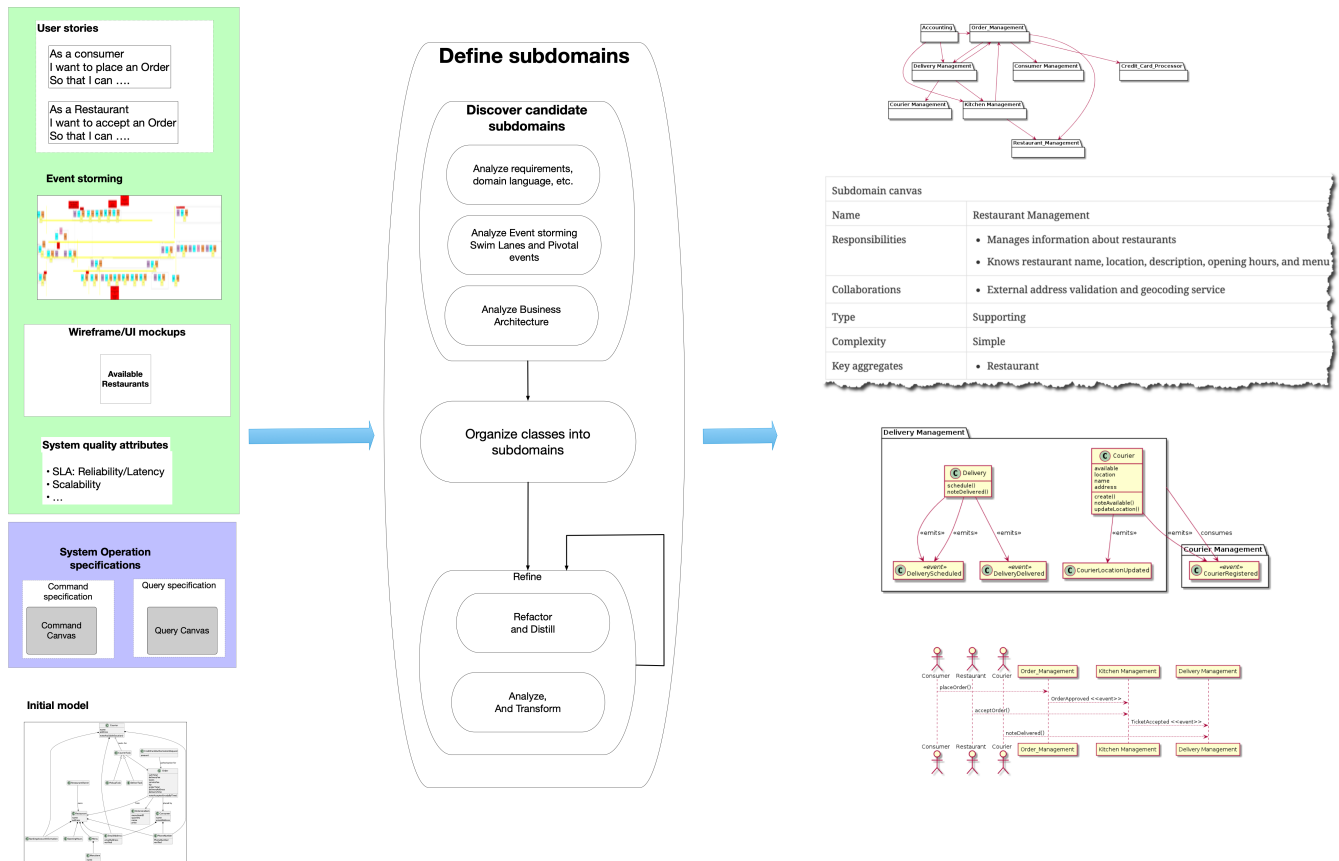
```
● Chriss-iMac-Pro:templates cer$ unzip -l microservice-architecture-design-templates-082022.zip '*01-discovering-system-operations*'
Archive:  microservice-architecture-design-templates-082022.zip
```

Length	Date	Time	Name
166460	08-19-2022	21:13	pdfs/01-discovering-system-operations.pdf
107571	08-19-2022	21:45	docx/01-discovering-system-operations.docx
3362	08-19-2022	21:02	01-discovering-system-operations.adoc
277393			3 files

Useful templates

Chapter 4. Kata #2: Define subdomains

Define and document the subdomains using the templates in [02-defining-subdomains](#)

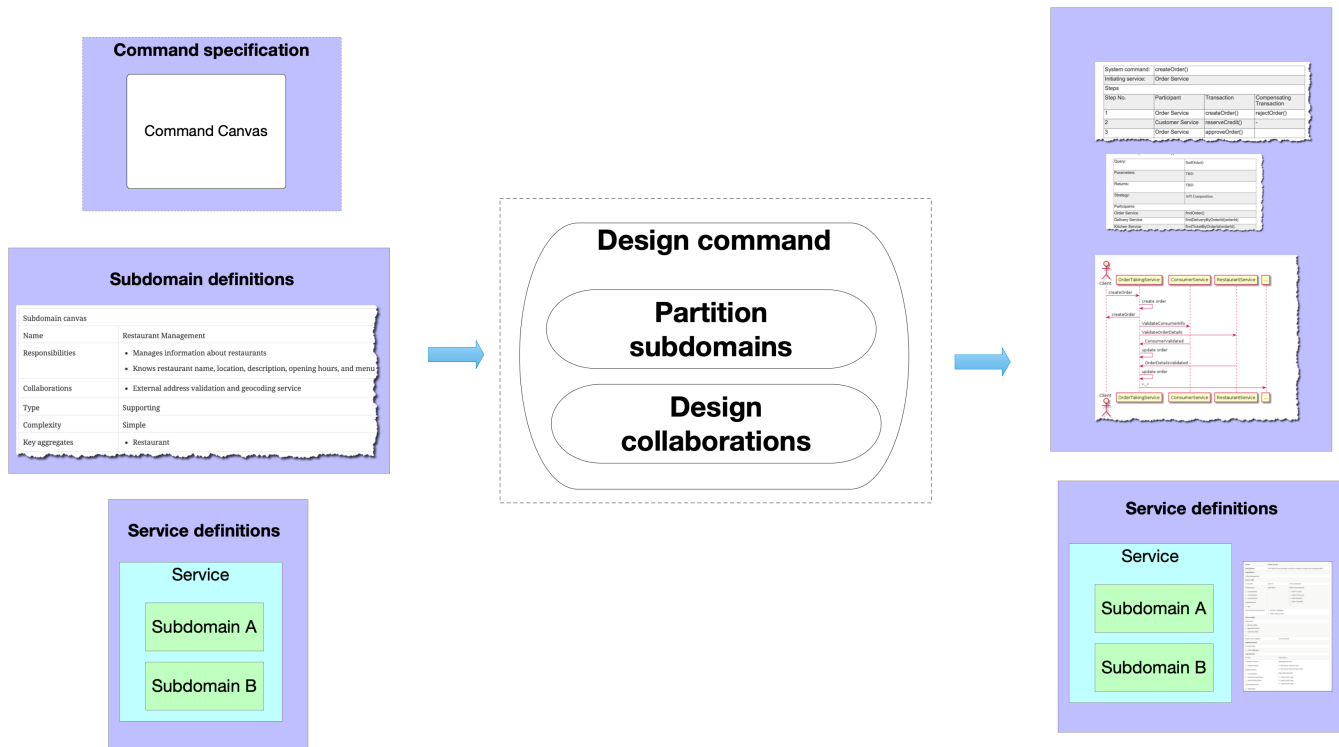


Chapter 5. Kata #3: Rank system operations

Rank the system operations using the template in 03-01-ranking-system-operations

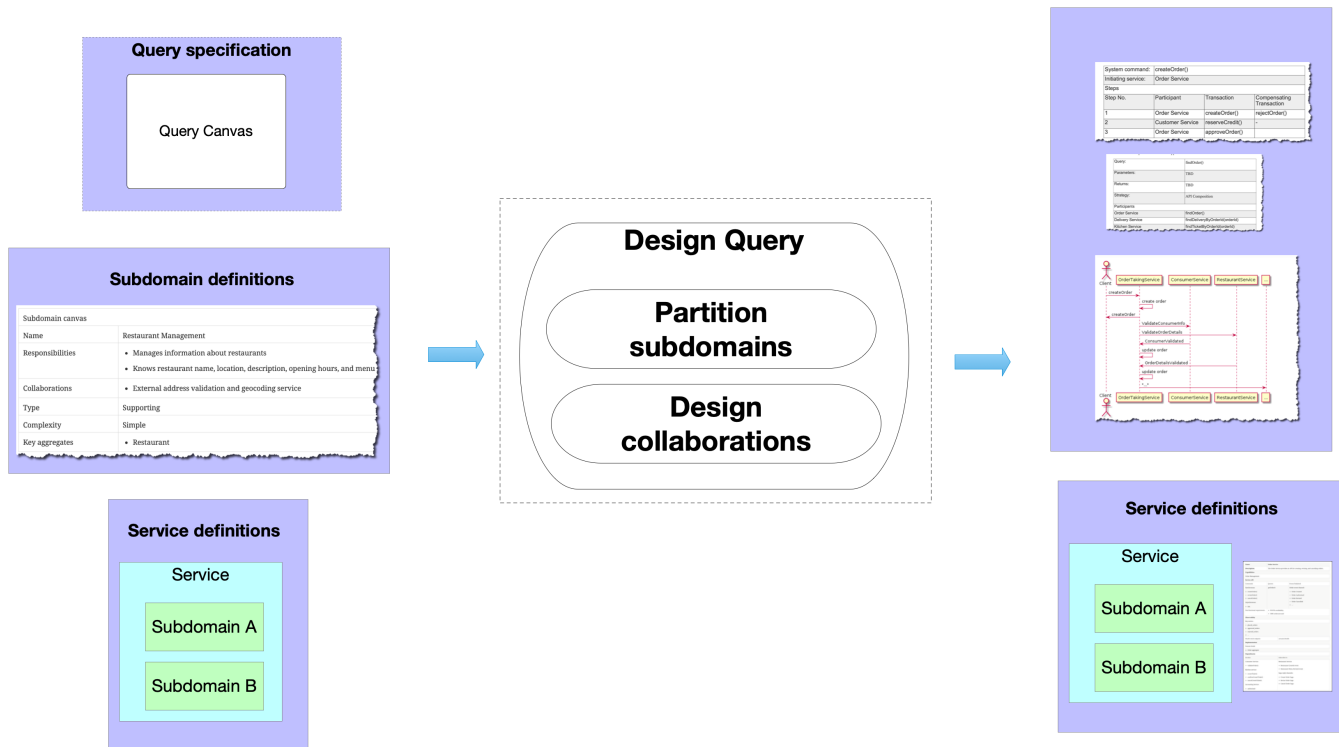
Chapter 6. Kata #4: Design system commands

Design the services and commands and document them using the templates in [03-02-designing-operations-and-services](#)



Chapter 7. Kata #5: Design system queries

Design the services and queries and document them using the templates in [03-02-designing-operations-and-services](#)



Chapter 8. Kata #6: Evaluate and refactor architecture

Steps:

1. Evaluate the architecture using the templates in 04-evaluating-a-microservice-architecture
2. Refactor the architecture to eliminate any smells and update the design documentation

